

Predicting Electrical Vehicle Charging Patterns at Public Charging Stations

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Abstract—As electrical vehicle (EV) technologies become mature, there is a rapid growth in public charging infrastructures. With public charging stations, an accurate prediction of local charging demand can enable many applications, such as dynamic charging allocations, optimizations of power grid operations, and proactive planning for EV users. However, charging demand prediction is a challenging problem, because it is usually affected by the diverse behaviors exhibited by different user groups. In this paper, we explore a data-driven approach to leverage historical charging records for the prediction of future charging demand. We develop a predictive model that distinguishes between the behaviors of both registered long-term users and unregistered short-term users. Utilizing a real-world dataset of 28053 records over 798 days at multiple locations, we employ several supervised learning algorithms to evaluate the performances of our models. Our evaluation results demonstrate that our model, enhanced with the XGBoost, significantly outperforms alternative solutions. It achieves a reduction in prediction error up to 40.8% at the finest time granularity (15-minute interval).

Keywords—EV charging availability prediction, user behaviors, machine learning

I. INTRODUCTION

As electric vehicles (EVs) quickly become popular with urban residents, more and more charging infrastructure has been deployed in public charging garages such as hospitals, malls, university campuses, etc. The charging records accumulated from these stations contain rich information about the spatiotemporal dynamics of local charging demand. Understanding the charging demand dynamics is very important for various applications: power grids can optimize power generation and distribution in response to anticipated demand spikes from fast-charging stations; charging station operators can allocate charging rates and adjust fees to better match fluctuating demand; and EV drivers can more effectively plan their routes based on the real-time availability of charging piles at nearby stations. Leveraging advanced analytics and predictive modeling can lead to a more robust energy management system, enhancing operational efficiency while promoting urban mobility and energy sustainability. Moreover, for EV users, precise behavior modeling is essential as it reliably predicts mobility needs, reduces range anxiety, and fosters greater integration of EVs into the grid [1]. Therefore, it is highly desirable and beneficial to accurately predict charging demand, as shown in our preliminary work [2].

The main challenge for EV charging demand prediction is that it is aggregated from individual users' charging behaviors,

which are affected by a number of factors such as battery status, work/travel schedule, parking policies and limitations, weather, etc. It is very difficult to obtain an accurate model for such a problem with a hidden high dimensionality. Moreover, the user base at charging stations is heterogeneous, encompassing both registered users and unregistered users. Registered users, who possess their unique IDs, typically tend to engage with the system over longer terms, e.g. months, or years; in contrast, unregistered users, lacking registered IDs, typically represent short-term or occasional visitors who use the charging stations sporadically. This diversity necessitates a predictive model that accommodates the varying patterns and frequencies of use exhibited by these two distinct user groups, ensuring that charging infrastructures can adapt dynamically to the needs of all users.

There is limited research on EV charging demand prediction, partly due to the scarcity of publicly available datasets [3] as most datasets are owned by commercial entities and remain inaccessible to the broader research community. A recent work [4] uses smart meter datasets to predict parking spot availability. In [5], a deep neural networks (DNN) is used to predict parking difficulty across different regions based on a user survey dataset. Furthermore, EV user behaviors have gradually attracted attention in the research community. There are lots of works studying charging scheduling algorithms and charging patterns [6], [7]. In [8], the authors modeled the state of EV charging as a Markov chain to solve the optimal power management for smart home microgrids, with the uncertainty of EV energy storage. The study [9] modeled the transition probability between driving and parking states at consecutive time intervals to capture the uncertainty of the arrival and departure time of each trip. These state variables, specifically 'driving' and 'not driving', are subsequently utilized in a charging scheduling scheme. However, the charging behaviors of various types of users have not been previously studied. Additionally, the available public EV charging datasets are typically very small, making it challenging to accurately identify and predict user behaviors.

To address these issues, we employ a data-driven approach to predict the usage patterns of EV charging poles at public stations. Typically, residential charging prediction is more predictable since residential users usually leave their EVs to charge overnight [3]. Unlike residential scenarios, for a real non-residential EV charging station, there are registered users

who regularly charge their EVs at this station and unregistered users who only charge their cars when occasionally traveling to this location. Hence, we design an ensemble predictive model that includes one predictive model for registered users and one for unregistered users. With a real-world charging record dataset [10], we segment it into subsets corresponding to unregistered and registered user records, organizing the data at various time granularity, such as 15-minute or 30-minute intervals. These subsets are then used to train supervised learning-based algorithms, specifically XGBoost [11], Support Vector Regression (SVR), and Gradient Boost Decision Tree (GBDT) to train these models. Based on these individual models, an ensemble predictive model is designed to generate the predicted output sequences that forecast the future availability of charging stations, providing a holistic view of anticipated charging patterns.

The primary contributions of this paper are as follows:

- To the best of our knowledge, we are the first to characterize the fine-grained charging behaviors of both long-term registered users and short-term unregistered users for public charging stations.
- To understand the various charging behaviors of different users, we design an ensemble predictive model for EV charging demand prediction of both registered users and unregistered users at workplace.
- We experimentally evaluated the performance of our model with a real-world dataset of 28053 records over 798 days at multiple stations. The results showed that our solution using XGBoost reduced prediction error by up to 40.8%, outperforming state-of-the-art solutions.

II. RELATED WORK

Demand pattern prediction in electric vehicles charging.

A substantial volume of research has been dedicated to predicting EV charging patterns, focusing on managing electricity demand and load in both residential and public areas. (i) *Residential charging*: For predicting residential charging demand, various ML methods have been employed to forecast the household day-head EV charging occurrence-time and ‘no-charge’ day respectively [12]. A study [13] investigated household charging demand, developing a novel framework that leverages historical data and user-specific features to train machine learning models, effectively predicting various charging characteristics, including connection and charging duration, as well as the demand and timing of subsequent sessions. (ii) *Public charging*: in public facilities, the prediction of charging demand has also attracted significant attention. To address the grid overloading due to unscheduled EV charging, ML models such as RF, LR, and XGBoost have been deployed to predict energy consumption patterns within the grid, enhancing energy management and preventing potential outages [14]. A proposed time-spatial forecast ML model EV charging-power demand at public fast-charging stations enhances adaptive charging infrastructures integrated with renewable resources and energy storage [15]. Additionally, the study [16] introduced a novel framework

that utilizes people’s travel trajectories to predict public charging demand, considering the supply and demand of transportation and the spatial-temporal patterns of EV user charging behaviors. Compared to these predictions, we predict the availability of charging piles at the non-residential charging station in order to guide EV users from a different viewpoint.

Behavior pattern prediction in electric vehicles charging.

EV charging behaviors can be discerned from detailed charging profiles or deduced from traveling and parking activities, revealing distinct patterns in terms of time, location, and energy consumption [17]. Recent advancements in predicting EV charging behaviors have introduced a variety of ML models. These techniques cover a broad range of approaches including support vector machine (SVM) [18], k-nearest neighbor (KNN) [19], LR [20], RF [21], XGBoost [22]. Particularly, the authors used KNN to find clusters of EV charging behavior based on real-world datasets, thus minimizing the influence of behavior uncertainties [19]. In [20], an optimization algorithm has been proposed to solve each EV charging station based on user charging behavior prediction by LR, satisfying the energy demand of individual drivers and improving the electric load profiles within a microgrid. The authors used XGBoost models trained on historical data to predict charging behaviors prediction [22]. Although these models are effective, many do not address the distinct behaviors of different user types, such as registered versus unregistered users, nor do they focus on characterizing fine-grained charging behaviors based on time granularity. From the methodological perspective, we make use of XGBoost, which has excellent performance on prediction, as our method to train the historical EV charging data, aiming to capture a more detailed and user-specific understanding of charging behaviors.

Machine learning with small datasets. Small datasets, distinct from large-scale datasets, present unique challenges in machine learning. While many successful ML applications rely on big data, effectively applying ML techniques to small datasets remains a critical task. For *unlabeled data*, transfer learning is a powerful approach, where knowledge gained from training on labeled data in one task is leveraged to solve related, yet distinct tasks [23]. For *imbalanced data*, techniques like convolutional neural networks (CNNs) are adapted with strategies such as class re-sampling or cost-sensitive training to address the skewness in class distribution [24]. For *insufficient data*, where only a few samples per class are available, are tackled using few-shot learning, which focuses on effective classification with minimal examples [25]. Compared to other traffic data types, the size of electric vehicle (EV) data is often smaller, particularly in terms of historical traffic volume. To address these challenges, MLCV [26] has been developed to estimate the trip energy consumption of EVs where the distribution of the training and test sets are inconsistent

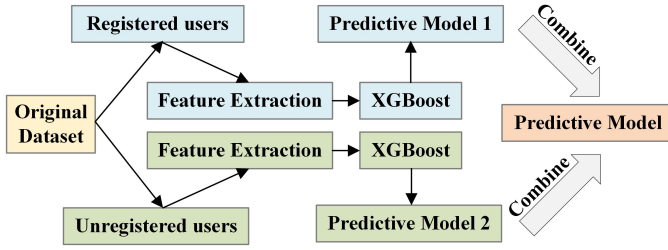


Fig. 1. An ensemble predictive model

due to the limited user driving behavior data [26]. Different from these works, our design utilizes an ensemble predictive model that combines predictors trained with different features.

III. EV CHARGING DEMAND PREDICTION MODELS

In this work, our primary goal is to predict the charging availability in a future time in order to help users grasp information to decide when they could charge their EVs. The availability is the integer of available charging piles at non-residential areas (workplaces).

Goal. Obtain a function f that inputs a sequence of the occupied number of charging piles in a previous continuous time period $OP_{TG,M} = \{o_1, o_2, \dots, o_m\}$ and then outputs a sequence of the available number of charging piles in a future continuous time period $AP_{TG,N} = \{a_1, a_2, \dots, a_n\}$. Here, TG presents time granularity. The element in $OP_{TG,M}$ and $AP_{TG,N}$ respectively mean the occupied and available number of charging piles at the time divided by TG . Besides, M and N present the number of TG .

$$f(OP_{TG,M}) = AP_{TG,N} \quad (1)$$

In our dataset, the total number of charging piles is constant, so the available number of charging piles is the total number minus the occupied number of charging piles. Considering the different charging patterns of registered EV users and unregistered EV users, we divide our function into two parts. So the improved function is as follows.

$$f(OP_{TG,M}) = f_{re}(OP_{TG,M}^R) + f_{unre}(OP_{TG,M}^U) \quad (2)$$

$OP_{TG,M}^R$ and $OP_{TG,M}^U$ are the sequences of the inputted historical number of registered users and unregistered users respectively.

Method. Based on the composition of the proposed target function and XGBoost, the input is a sequence of the occupied number of charging piles in a previous continuous time period, and the output is a sequence of the available number of charging piles in a future continuous time period. As Fig. 1 shows, for registered users and unregistered users, we use XGBoost to train two predictive models separately and then combine the middle prediction results to obtain the final prediction results.

Data processing. Based on the individual charging pile at the charging station, we would calculate the availability (0: unavailability, 1: availability) of each charging pile in a certain historical period. Then, combining these results would obtain the total occupied and available number of charging piles in this period. For ML methods, there is no need to separate weekdays and weekends to train the models because they can use data of a continuous-time sequence.

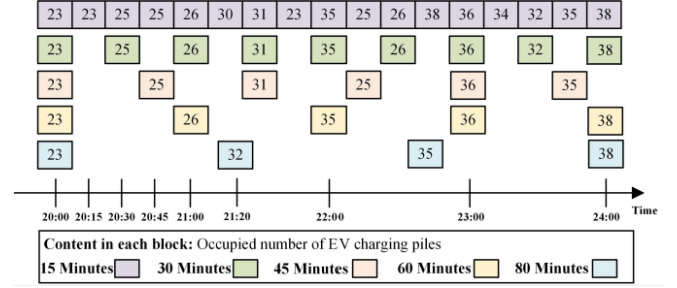


Fig. 2. Data splitting of unregistered/registered users by different time granularities

Fig. 2 shows an example of data splitting of unregistered and registered users by different time granularities. In order to predict fine-grained, we divide the time of a day into 15, 30, 45, 60, 80 minutes. The reason for choosing 80-minute instead of 75-minute is that the former obtains integral parts of a single day. We could see that from 20:00 to 24:00 of a day, in each time granularity, there are different quantities of split data.

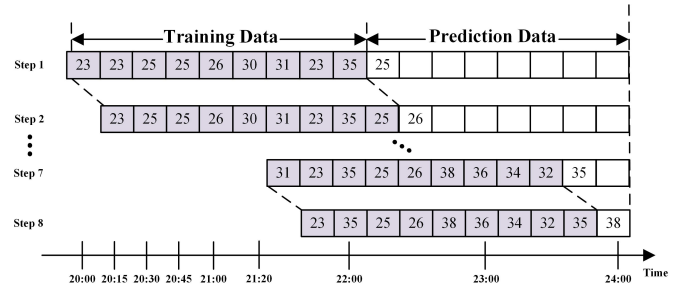


Fig. 3. Prediction process when time granularity is 15-minute

Prediction process. To predict $AP_{TG,S-M} = \{a_{m+1}, a_{m+2}, \dots, a_s\}$, a sequence of the available number of charging piles in past time, we need to use $OP_{TG,M} = \{o_1, o_2, \dots, o_m\}$, a sequence of the occupied number of charging piles in future time. First, use $\{o_1, o_2, \dots, o_m\}$ to fit a function and get $\{a_{m+1}\}$. Then, use $\{o_2, o_3, \dots, a_{m+1}\}$ records to fit a function and get $\{a_{m+2}\}$ record. Conduct this multi-step iterative process until obtaining $\{a_s\}$ record. To obtain accurate results, there is an accuracy threshold to guarantee the accuracy because, at each prediction step, we add the prediction data of the last step to the training data, thus increasing the possibility of a loss of accuracy. Therefore, if the result at one certain step has not achieved requirements, the prediction length cannot

TABLE I
SELECTED FEATURES IN EV CHARGING PATTERN PREDICTION

Name&Description	Example
connectionTime(time when plugged in)	23 May 2018 20:04:39
disconnectTime(time when unplugged)	23 May 2018 23:08:50
spaceID(Identifier of a charging pile)	CA-495
userID(Identifier of a user)	68

be increased, which means the prediction process is finished and we cannot get $\{a_s\}$ record.

Fig. 3 shows a specific prediction process when the time granularity is 15-minute. In this case, suppose that the initial sequence of available number of charging piles from 20:00 to 24:00. At step 1, the training data is $\{23, 23, 25, 26, 30, 31, 23\}$ and the prediction data is $\{35\}$. At step 2, we add $\{35\}$ to the end of the initial sequence and discard the first record $\{23\}$ in the initial sequence, then train the model. Repeat these steps, finally, at step 8, the training data is $\{23, 35, 25, 36, 38, 36, 34, 32, 35\}$ and the prediction data is $\{38\}$. Hence, we get the last prediction result. The prediction process for registered users and unregistered is the same, and the difference is the split dataset we used.

Prediction results. As Fig. 1 shows, for registered users, we could use predictive model 1 to get prediction results. Here, the prediction result for registered users with data splitting by TG is

$$AP_{TG,N}^R = \{R_1, R_2, \dots, R_N\} \quad (3)$$

The predictive model 2 is used to obtain prediction results for unregistered users, and the prediction result is

$$AP_{TG,N}^U = \{U_1, U_2, \dots, U_N\} \quad (4)$$

In order to ensure obtaining reasonable results, when the prediction accuracy of one of the predictive models does not reach the accuracy threshold at a certain step, these two predictive models would finish their training process at the same time, thus obtaining prediction results with the same length of the time interval. Finally, by combining these two prediction results, the available number of EV charging piles in a future time period is

$$\begin{aligned} AP_{TG} &= AP_{TG,N}^R + AP_{TG,N}^U \\ &= \{R_1 + U_1, R_2 + U_2, \dots, R_N + U_N\} \end{aligned} \quad (5)$$

Here, AP_{TG} represents the final aggregate prediction of available charging piles.

IV. EXPERIMENTS

In this section, we evaluate our proposed prediction model with different machine learning methods, specifically XGBoost, GBDT, and SVR, the primitive model without considering user behaviors is also trained with XGBoost as the baseline algorithm.

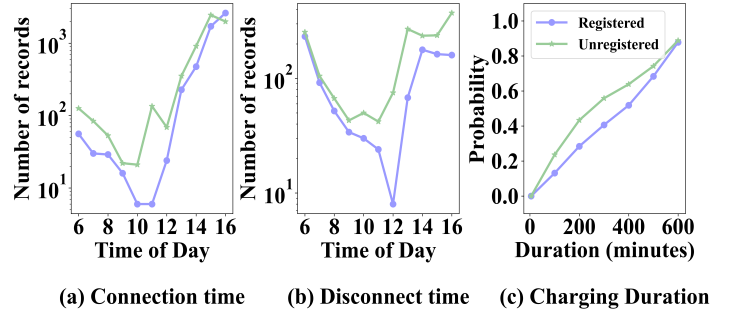


Fig. 4. Distribution of dis/connection time and CDF of charging duration of sections between different users

Dataset. The dataset we use contains a total of 798 days from April 2018 to July 2020 [10], and there are 52 charging piles in this one charging station. The selected features are shown in Table I. For unregistered users, there are 14768 records, and for registered users, there are 13285 records.

Metrics. To evaluate the accuracy of predicted results, two main quantitative metrics are utilized. The first metric is the Root Mean Squared Error (RMSE), a commonly used measure of the differences between values predicted by a model and the values actually observed. It is defined as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (6)$$

where y_i represents the observed number of EV users for the i -th sample and $\hat{y}_i$ is the estimated number. Generally, a lower RMSE indicates better predictive accuracy. However, comparing RMSE values across different datasets can be misleading due to the dependency of RMSE on the scale of the data. The second metric is the Mean Absolute Percentage Error (MAPE), which measures the average magnitude of the errors in percentage terms, thus providing a scale-independent measure of prediction accuracy. It is defined as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \times 100\% \quad (7)$$

In this case, a smaller MAPE is preferable, as it indicates a lower error rate in percentage terms.

Model comparison. Here's the comparison of the models we use in the following experiments: XGBoost, GBDT, and SVM are three prevalent models used in EV prediction [3]. GBDT, which integrates multiple Decision Trees to create a robust classifier, enhances accuracy by compensating for individual trees' errors, but it can overfit and is sensitive to dataset noise [27]. In contrast, XGBoost extends GBDT's capabilities by incorporating advanced regularization and improving computational efficiency, making it particularly effective for small datasets [11]. On the other hand, SVM demands meticulous kernel selection, a critical yet challenging process that often renders it less suitable for rapid deployment [28]. Unlike SVM, which requires extensive tuning in high-dimensional spaces,

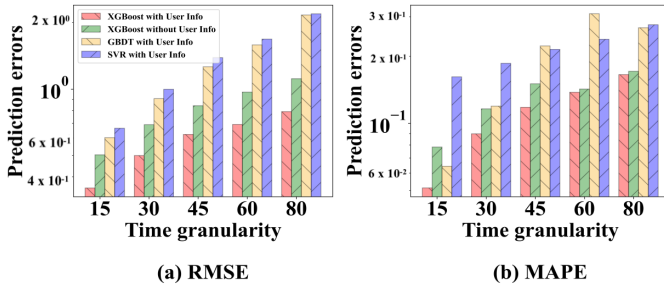


Fig. 5. Prediction errors of primitive models with XGBoost, GBDT, and SVR with different time granularity

XGBoost and similar tree ensembles are easier to configure and generally produce robust results across various data scenarios [29]. Deep learning based models are not selected as they typically don't work well with small datasets. These models require significantly more training data and often suffer from overfitting, which reduces their generalizability and transferability [30].

User data analysis. The distribution of charging connection time and disconnection time between these two types of users is shown in Fig. 4 (a) and (b). Here we select the time from 6:00 to 16:00 and we can see that there are significant differences in the daytime. The CDF (cumulative distribution function) of charging duration of sections between two types of users is shown in Fig. 4 (c). Here we select the duration from 0 to 600 minutes and it shows when the duration is shorter, the CDF of unregistered users is larger than that of registered users, which means the majority of unregistered users have shorter durations. In the divided dataset of each time granularity, we used 75 percent of the data as training samples and 25 percent as testing samples.

An ablation study on user ID. We compared the proposed prediction model with XGBoost and the primitive model with XGBoost to show the effect of considering user information. In the primitive model with XGBoost, the availability of charging piles has been predicted in a future time period based on previous records with no distinction between users. However, in the proposed prediction model with XGBoost, the future availability of charging piles comes from previous registered and unregistered user records. In our dataset, there are many true values which equal 0, so when calculating MAPE, we used a method called Laplace Smoothing, adding 1 to each value for avoiding mistakes [31].

By using the XGBoost method, the comparisons between the primitive model with XGBoost and the proposed prediction model with XGBoost have been shown in Fig. 5. We could see that the performance of both RMSE and MAPE in our proposed prediction model with XGBoost is better than the primitive model with XGBoost in each time granularity. Especially, when the time granularity is 15-minute, the difference between these two is significant. Here, we chose the numbers of time granularity when each method reaches its minimum.

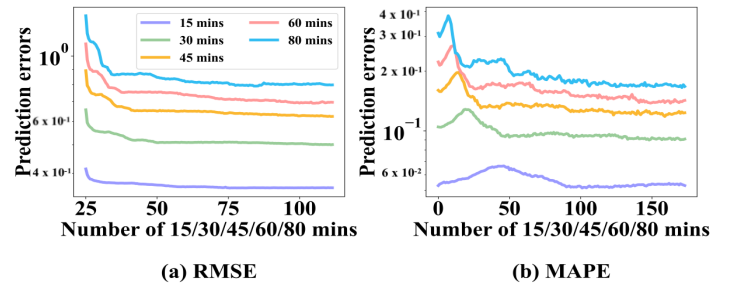


Fig. 6. Prediction accuracy v.s. size of training dataset

Comparison under different time granularity. We chose SVR and GBDT to predict charging availability in our proposed model, compared with XGBoost. As Fig. 5 (a) shows, for RMSE, the proposed prediction model with XGBoost performs the best in each time granularity and the proposed prediction model with SVR performs the worst. For MAPE, as Fig. 5 (b) shows, the proposed prediction model with XGBoost also performs the best in each time granularity. The proposed prediction model with GBDT performs better than with SVR when time granularity is 15-minute, 30-minute, and 80-minute, while the proposed prediction model with SVR performs better than the proposed prediction model with GBDT when time granularity is 45-minute and 60-minute.

In the proposed prediction model with XGBoost, we try to find the better scheme, including which time granularity and what length of training data in different time granularities is better, to obtain more reasonable results.

As Fig. 6 shown, the smaller the time granularity, the better the performances of RMSE and MAPE. It shows 15-minute granularity is a priority method when predicting charging availability in our dataset. Besides, the reasonable number of 15-minute time granularity, which means the length of training data in the model, is in the interval between 25 and 50. We could also see that in each time granularity, RMSE appears a downward trend from the beginning, while MAPE firstly increases and then decreases as the number of time granularity increases.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed a data-driven approach for predicting EV charging availability at workplaces by considering user information and behaviors. Experiments on ACN-Data showed that the accuracy of our model trained with XGBoost outperforms the other machine learning solutions. This work reveals that considering user-specific charging behaviors in charging demand predictive models can significantly improve the model accuracy. A more comprehensive study is needed to understand the spatiotemporal charging patterns and their context embedded in the data. Also, advanced machine learning methods such as Sequential & Contextual Neural Network and Time-Inhomogeneous Markov Chain can be employed to perform time series prediction.

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